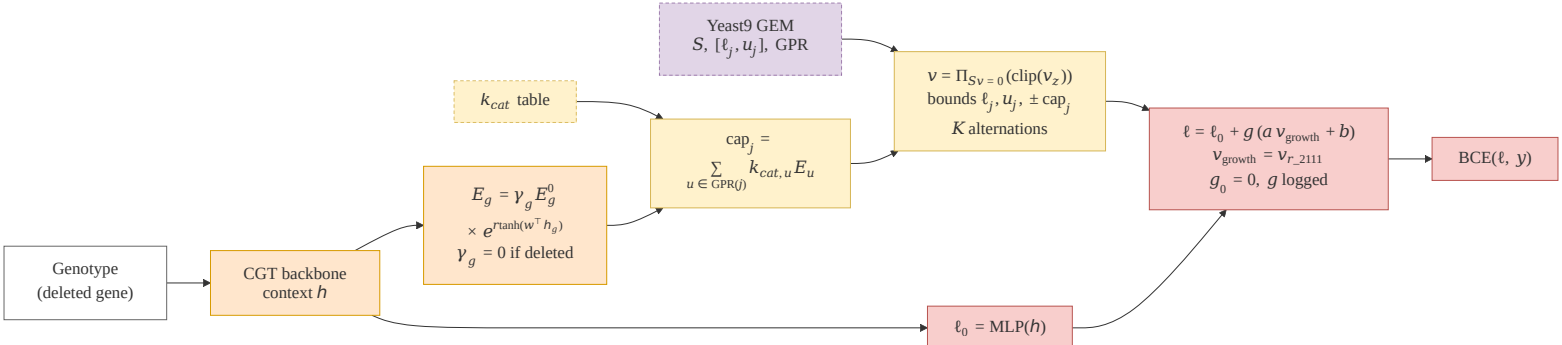

Genotype
(deleted gene)
CGT backbone
context h
$E_g = \gamma_g E_g^0$
$\times e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ if deleted
k_{cat} table
Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} k_{cat,u} E_u$
$v = \Pi_{Sv=0}(\text{clip}(v_z))$
bounds $\ell_j, u_j, \pm \text{cap}_j$
K alternations
$\ell_0 = \text{MLP}(h)$
$\ell = \ell_0 + g\,(a\, v_{\text{growth}} + b)$
$v_{\text{growth}} = v_{r_2111}$
$g_0 = 0$, g logged
BCE(ℓ, y)